

BAYES PREDICTIVE DENSITY FUNCTION FOR THE ACCELERATED LIFE TESTS VIA ORTHOGONAL PARAMETERS

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Summary: Accelerated life testing of an item, under more severe than normal conditions, is commonly used by industrial statisticians to reduce test time and costs. In this paper, the Bayes predictive density function via orthogonal parameters for a future random time under the usual stress level is considered. A power rule model and an exponential distribution with the type II censoring mechanism are assumed. Some examples are considered to illustrate the performance of the Bayes predictive density.

1. INTRODUCTION

In engineering applications, accelerated conditions are produced by testing items at higher than normal temperature, voltage, pressure, load, etc. (see Mann, Shaffer & Singpurwalla (1974); Nelson (1990); Achcar (1990, 1991) and Proshan & Singpurwalla (1979)). In this paper, we discuss from the Bayes point of view the construction of the predictive density function for a future random time under the exponential censored data. This predictive density is obtained via the orthogonal parameters, that is, the Fisher information matrix is diagonal (see Cox & Reid, (1987)). The main advantage of this orthogonality is to obtain a known predictive density and a closed form for the predictor of a future random time. We assume an exponential model with a power rule model and a type II censored data. A valuable survey of accelerated life testing problems from the Bayesian perspective is given by Mazzuchi and Singpurwalla (1986).

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2. FORMULATION OF THE MODEL

2.1. INTRODUCTION

We will consider the following model: Suppose that we perform life tests on a set of n_i devices under the stress V_i and the life time T has an exponential density

$$(1) \quad f(t | \theta_i) = \frac{1}{\theta_i} \exp \left\{ -\frac{t}{\theta_i} \right\}, \quad \theta_i > 0, \quad t > 0,$$

$$i = 1, \dots, k.$$

The unknown parameter θ_i , $i = 1, \dots, k$, is the mean time to failure under the stress V_i . The k tests yield, as data, the set $\{V_i, n_i, \theta_i\}$, $i = 2, \dots, k$, where $\theta_i = \frac{A_i}{r_i}$, $A_i = \sum_{j=1}^{r_i} t_{ij} + (n_i - r_i)t_{r_i+1}$ and t_{ij} , $j = 1, \dots, r_i$ are observable failures under the stress V_i for n_i items on test ($r_i < n_i$). Given the data set and the usual stress level V_1 , our purpose is to get information of the mean time θ_1 . To make inference on θ_1 , we adopt the power rule model (see for example, Mann, Schaffer & Singpurwalla, 1974).

$$(2) \quad \theta_i = \frac{\alpha}{V_i^\beta}, \quad \alpha > 0, -\infty < \beta < \infty, \quad i = 1, \dots, k.$$

The log-likelihood function for θ_1 and β is given by

$$(3) \quad \ell(\theta_1, \beta) = -r \ln(\theta_1) + \beta r \ln(\bar{V}) - \frac{1}{\theta_1} \sum_{i=2}^k A_i \left[\frac{V_i}{V_1} \right]^\beta, \quad \text{where}$$

$$\bar{V} = \prod_{i=2}^k \left[\frac{V_i}{V_1} \right]^{\frac{r_i}{r}} \quad \text{and} \quad r = \sum_{i=2}^k r_i.$$

First, we estimate β in the presence of the nuisance parameter θ_1 . The procedure can be simplified by transforming the origin parameters (θ_1, β) to

the orthogonal parameters (λ, β) . The orthogonal parameter λ is obtained by solving the differential equation (Cox & Reid, 1987)

$$(4) \quad \frac{\partial \theta_1}{\partial \lambda} = \ln(V) \frac{\partial \theta_1}{\partial \beta}, \text{ giving}$$

$$(5) \quad \lambda = \frac{\theta_1}{\beta V} \quad (6)$$

Then the new likelihood function in terms of (λ, β) is

$$(6) \quad \ell(\lambda, \beta) = -r \ln(\lambda) - \frac{1}{\lambda} \sum_{i=2}^k A_i \left[\frac{V_i}{V V_1} \right]^{\beta}$$

It is easy to verify that $\hat{\beta}$, the estimator of β given λ , is the solution of

$$(7) \quad \frac{\partial \ell(\lambda, \beta)}{\partial \beta} = \sum_{i=2}^k A_i \left[\frac{V_i}{V V_1} \right]^{\beta} \ln \left[\frac{V_i}{V V_1} \right] = 0$$

and that the Fisher information matrix for (λ, β) is

$$(8) \quad I(\lambda, \beta) = \begin{bmatrix} \frac{r}{\lambda^2} & 0 \\ 0 & \sum_{i=2}^k r_i \ln^2 \left[\frac{V_i}{V V_1} \right] \end{bmatrix}$$

Note that $\hat{\beta}$ is obtained independently of λ , that is, $\hat{\beta}$ is stable with respect to λ . The Jeffrey's prior for λ and β (Box & Tiao, 1973, p. 54) is given by

$$\begin{aligned}\pi(\lambda, \beta) &\propto \{\text{deter } I(\lambda, \beta)\}^{\frac{1}{2}} \\ &= \frac{1}{\lambda} \left[r \sum_{i=2}^k r_i \ln^2 \left[\frac{V_i}{V_1} \right] \right]^{\frac{1}{2}}.\end{aligned}$$

That is,

$$(9) \quad \pi(\lambda, \beta) \propto \frac{1}{\lambda}.$$

For fixed λ , expanding $\ell(\lambda, \beta)$ in a second-order Taylor series about the maximizing value $\hat{\beta}$ of $\ell(\lambda, \beta)$, we have the Laplace approximation (Tierney & Kadane, 1986) for the marginal posterior of λ given by

$$(10) \quad \pi(\lambda | \text{data}) \propto \lambda^{-(r+\frac{1}{2})} \exp \left\{ -\frac{1}{\lambda} \sum_{i=2}^k A_i \left[\frac{V_i}{V_1} \right]^{\hat{\beta}} \right\}.$$

Taking $\beta = \hat{\beta}$, we have from (5) and (10) the marginal posterior of θ_1 given by

$$(11) \quad \pi(\theta_1 | \text{data}) \propto \theta_1^{-(r+\frac{1}{2})} \exp \left\{ -\frac{1}{\theta_1} \sum_{i=2}^k A_i \left[\frac{V_i}{V_1} \right]^{\hat{\beta}} \right\}.$$

3. PREDICTIVE DENSITY FUNCTION FOR A FUTURE RANDOM TIME UNDER THE USUAL STRESS LEVEL V_1

The Bayes predictive density for a future random time $T_{n+1,1}$, under the usual stress V_1 and $n = \sum_{i=2}^k n_i$, is defined as

$$p(t_{n+1,1} | \text{data}) = E_{\pi}[f(t_{n+1,1} | \theta_1)]$$

where the expectation is with respect to (11). For more details of this definition, see for example, Zellner, 1975, p. 29. The Bayes-predictive density is very important in density estimation problems and quality control. After some easy computations we obtain that

$$(12) \quad p(t_{n+1,1} | \text{data}) = \frac{V_1^\beta \left\{ \sum_{i=2}^k A_i V_i^\beta \right\}^{r-\frac{1}{2}}}{\left\{ \sum_{i=2}^k A_i V_i^\beta + V_1^\beta t_{n+1,1} \right\}^{r+\frac{1}{2}}}$$

$t_{n+1,1} > 0$, (Pareto density).

The mean and the variance of the predictive density of $T_{n+1,1}$ are

$$(13) \quad E[T_{n+1,1} | \text{data}] = \frac{\sum_{i=2}^k r_i \theta_i \left\{ \frac{V_i}{V_1} \right\}^\beta}{r - \frac{3}{2}} \quad \text{and}$$

$$(14) \quad V[T_{n+1,1} | \text{data}] = \frac{r - \frac{1}{2}}{r - \frac{5}{2}} \left\{ \frac{\sum_{i=2}^k r_i \theta_i \left\{ \frac{V_i}{V_1} \right\}^\beta}{r - \frac{3}{2}} \right\}^2,$$

respectively.

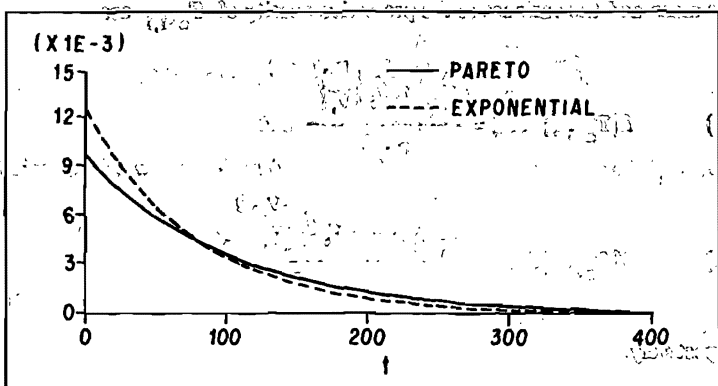
4. SOME EXAMPLES

Example 5.1

Consider the data of table 1 which was generated according to (1) and power rule model (2) for 5 stress with $\alpha = 500$ and $\beta = 0.8$.

TABLE 1. Generated data with $\alpha = 500$ and $\beta = 0,8$.

i	V_i	θ_i	n_i	r_i	A_i	Uncensored observations
1	10	79,24				
2	20	45,52	30	8	502,5	2,5 4,7 8,0 9,0 9,7 14,2 17,3 18,9
3	30	32,90	30	12	385,2	0,7 1,2 5,6 5,7 7,0 8,5 10,0 11,1 13,3 14,9 15,2 15,3
4	40	26,14	30	18	283,9	0,7 1,5 1,7 2,3 2,4 2,7 4,6 5,4 5,8 7,4 7,6 8,1 8,9 10,5 11,3 12,8 13,3 13,5
5	50	21,87	30	22	555,6	1,9 2,8 3,2 3,7 5,2 7,4 7,4 7,8 9,9 10,9 11,4 12,0 12,7 13,6 14,5 15,1 16,7 19,0 20,3 27,2 30,3 33,5

Figure 1. Bayes predictive density and exponential density for $\theta_i = 79,24$.

In this simulation we obtain:

$$\hat{\beta} = 0,9931 \quad \text{and}$$

$$E[T_{1:n,1} | \text{data}] = 102,90.$$

We see from figure 1 that the Bayes predictive density is similar to the exponential density for the future observation $T_{202,1}$ under the usual stress $V_1 = 10$ and $\theta_1 = 79,24$.

Example 5.2

In table 2 we give the summary of the data set introduced by Barlow, Toland and Freeman (1986). This data set consists of lifetimes of Kevlar 49/epoxy spherical pressure vessels that are subjected to constant sustained pressure until vessel failure, commonly known as static fatigue or stress-rupture. Barlow, Toland and Freeman (1986) verified that the use of an exponential distribution for this data is appropriate under 80% and 90% stress levels. Because complete data sets were available for these stress levels, we assume that $n_i = r_i$. We also assume the power rule model in (2) and that the usual stress level is $V_1 = 75\%$.

TABLE 2. Summary of the Kevlar 49/epoxy strands at 80% and 90% stress levels

i	V_i (%)	$n_i = r_i$	A_i
1	75		
2	80	100	209,00
3	90	101	104,33

From (13) and (7) we have that

$$\hat{\beta} = 45,106 \text{ and}$$

$$E[T_{202,1} | \text{data}] = 3869,6872.$$

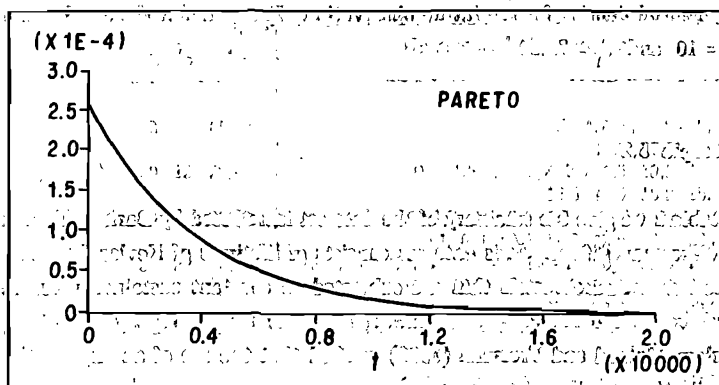


Figure 2. Bayes predictive density for $T_{202,1}$

Figure 2 gives the performance of a future observation $T_{202,1}$ for the Kevlar 49/epoxy strands at 75% stress level.

5. CONCLUSIONS

Using the orthogonal parameterization and Laplace approximation it was possible to get a closed form for the Bayes predictive density for a future random time $T_{n+1,1}$ under the usual stress V_1 . The results (12), (13) and (14) could be of great practical interest because we do not need expensive numerical integration to get them. The simulation study illustrates the good performance of the Bayes predictive density.

6. ACKNOWLEDGEMENT

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7. REFERENCES

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